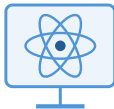


Multi-fidelity problem

self-driving laboratory

candidate pool $\mathcal{X}_{\text{cand}}$

13 benchmarks: 4 synthetic · 9 chemistry & materials



Low fidelity

simulation
cheap, abundant



High fidelity

experiment
costly, scarce



Controlled comparison

fix budget · schedule · capacity

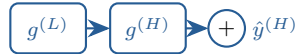
→ **vary only the surrogate**

**Gaussian-process
(GP) family ×3**



GP-base · GP-DKL
GP-SV

**Transfer-learning
(TL) surrogates ×5**



residual transfer + Bayesian linear-regression head



Transfer-learning wins

on chemistry & materials benchmarks

Better solutions

regret vs. budget



TL-base $\geq$ GP-base on all 9 chemistry & materials pools

Faster

compute per optimization loop

GP

TL **up to 22× fewer FLOPs**